Digital Assignment

TransferWorth: Cross-League Performance Translation Model (CLPTM)

A Machine Learning & Explainable AI Platform for Cross-League Football Recruitment

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1. Abstract

In professional association football, player recruitment represents one of the highest-stakes financial and sporting investments, with European clubs committing billions of euros annually during transfer windows. A chronic and expensive failure mode in football recruitment is the breakdown of metric translation across leagues: a forward scoring 0.90 goals per 90 minutes in one domestic league frequently suffers severe performance drop-offs when transferring to another. This disconnect—exemplified by the 'Haaland vs. Werner' paradox—stems from non-linear disparities in league tempo, tactical pressing structures, space compactness, teammate creation quality, and physiological adaptation demands.

To address this challenge, we present TransferWorth (Cross-League Performance Translation Model - CLPTM), an institutional-grade, full-stack machine learning system that translates, projects, and contextualizes professional football player performance across Europe's premier competitions. The system ingests and harmonizes a multi-season dataset comprising 24,413 player-seasons (14,865 meeting the professional playing threshold of ≥ 600 minutes) across Europe's Top 5 domestic leagues (Premier League, La Liga, Serie A, Bundesliga, and Ligue 1) spanning 2017/18 through 2025/26.

TransferWorth employs a leak-free multi-model architecture combining LightGBM Gradient Boosted Decision Trees for primary metric regression, Deep Multilayer Perceptrons (MLP) for non-linear tactical interactions, and Bayesian Ridge Regression to compute explicit 95% Bayesian Credible/Confidence Intervals for projected metrics. Furthermore, the model introduces a Tactical Transfer Reality Check (TRC) engine that modulates predictions based on the target club's relative attacking strength (Δstrength ratio), granting up to +10% performance mitigation for dominant clubs or applying up to -12% penalties for struggling sides. Supervised adaptation and downside risk classifiers, calibrated on 955 empirical cross-league transfers, estimate the probability of output retention (ROC-AUC: 0.611) and flag severe collapse risks (ROC-AUC: 0.655).

Explainable AI is integrated via TreeSHAP factor attributions and interactive counterfactual what-if simulations. Rigorous backtesting on 550 longitudinal transfer cases confirms predictive accuracy: Mean Absolute Error (MAE) of 0.0545 on goals_p90, 0.0333 on assists_p90, 0.0359 on xg_p90, and 0.0127 on xag_p90. The system is deployed via a normalized 8-table relational database schema, a high-throughput FastAPI REST microservice with JWT authentication, and a responsive React 19 / Vite analytical dashboard.

> **Performance Summary**
> *Key Quantitative Result: Across 550 verified longitudinal transfer test cases, the primary LightGBM model achieved an out-of-sample MAE of 0.0545 on goals_p90 and 0.0127 on xag_p90, outperforming conventional league multipliers and standard linear regression baselines.*

2. Problem Statement

2.1 The Economic Stakes of Player Recruitment

Modern football has evolved into a globalized, hyper-commercialized industry where single player transfers routinely command fees exceeding €50 million to €100 million, alongside long-term multi-million-euro wage commitments. FIFA transfer market reports indicate that clubs spend over €8 billion annually on international player movements. Consequently, an erroneous transfer acquisition inflicts not only catastrophic financial losses

(depreciated asset valuation, deadweight wage burdens) but also sporting setbacks that can result in missed UEFA Champions League qualifications or relegation.

2.2 The Metric Carry-Over Fallacy & The 'Haaland vs. Werner' Paradox

Recruitment departments historically relied on unadjusted statistical carry-over—naively projecting that a player's raw per-90 metrics in League A will directly replicate in League B. This assumption repeatedly fails due to profound environmental and tactical divergence between European leagues:

- **The 'Haaland vs. Werner' Paradox:** In the German Bundesliga, Erling Haaland (Borussia Dortmund) and Timo Werner (RB Leipzig) both exhibited elite goal-scoring rates (~0.90 to 1.00 goals/90). Upon transferring to the English Premier League, their trajectories diverged drastically. Erling Haaland maintained record-breaking output at Manchester City, whereas Timo Werner experienced an immediate 60% decline in goal conversion and minutes security at Chelsea.
- **League Difficulty Deltas:** Competitions vary significantly in pace, physical dueling intensity, and defensive organization. Bundesliga games often feature open-field transitional play, while the Premier League and Italian Serie A feature high physical pressing compactness and deeper defensive low-blocks.
- **Teammate Creation Capacity:** A player moving to an average mid-table team will receive fewer quality progressive passes and box entries than one moving to a possession-dominant superclub (e.g., Manchester City, Real Madrid, Bayern Munich). Raw league multipliers ignore target club context entirely.
- **Tactical System Mismatch:** Transition-oriented forwards who excel when running into open channels behind high defensive lines struggle when tasked with breaking down compact low-block defenses in closed spaces.
- **Adaptation Uncertainty:** Standard analytics provide single-point projections, leaving decision-makers blind to epistemic uncertainty, downside risk, and acclimatization variance.

2.3 Shortcomings of Conventional Scouting Systems

Traditional scouting workflows suffer from three structural weaknesses:

- **1. Eye-Test & Recency Bias:** Scouts frequently overweight recent highlight-reel performances in international tournaments or domestic cup matches while discounting multi-season statistical variance.
- **2. Uniform League Multipliers:** Fixed league coefficients (e.g., multiplying all Bundesliga goals by 1.0 and Portuguese Primeira Liga goals by 0.75) treat all players uniformly, ignoring positional profiles, age curves, and target club quality.
- **3. Absence of Explainability & Uncertainty:** Black-box algorithmic scouting tools fail to communicate why a forecast is generated or what tactical conditions must be satisfied to unlock the player's potential.

3. Introduction

3.1 Domain Background

Association football analytics has transitioned over the past decade from rudimentary box-score counts (goals, assists, appearances) to granular event tracking. The public release and commercialization of advanced metrics—Expected Goals (xG), Expected Assisted Goals (xAG), Key Passes, Progressive Passes, Progressive Carries, and Defensive Pressures—has enabled analysts to quantify the underlying quality of chances created and conceded, moving beyond misleading outcome variance.

However, almost all state-of-the-art predictive models remain confined to intra-league forecasting (e.g., predicting how a player will perform next season within the same domestic competition). Modeling player performance across disparate domestic leagues requires solving a domain-adaptation and covariate-shift problem in applied machine learning: how to translate a feature vector observed under one environment distribution P(X_LeagueA) into an outcome distribution governed by another P(Y_LeagueB).

3.2 Genesis and Vision of TransferWorth (CLPTM)

Developed as an academic software engineering and machine learning project at the School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Chennai, under the leadership of Shashank Deodhar Bantupalli (Reg No: 24BRS1297), TransferWorth (Cross-League Performance Translation Model) was conceived to provide football directors, chief data officers, and technical scouts with an end-to-end analytical workbench.

The system combines rigorous data engineering, empirical football physics, machine learning regression, probabilistic Bayesian uncertainty estimation, game-theoretic explainability (SHAP), and modern cloud/web technologies into an integrated software platform.

4. Objectives and Scope

4.1 Functional Objectives

- **1. Multi-League Data Ingestion:** Ingest, parse, sanitize, and standardize historical match event statistics across Europe's Top 5 domestic leagues from 2017/18 through 2025/26 (24,413 records).
- **2. League Strength Quantification:** Calculate dynamic League Strength Coefficients (LSC) based on UEFA 5-year confederation coefficients and empirical cross-league transfer performance ratios.
- **3. Multi-Target Metric Forecasting:** Forecast translated per-90 metrics across attacking (goals, assists, xG, xAG), progression (prog passes, key passes, prog carries), and defensive categories.
- **4. Tactical TRC Context:** Incorporate destination team attacking strength ratios (Δstrength) to reward players moving into high-creation squads and penalize moves to struggling teams.
- **5. Uncertainty Modeling:** Deliver 95% Bayesian Confidence/Credible Intervals ([μ - 1.96σ, μ + 1.96σ]) on projected output.
- **6. Adaptation & Risk Classification:** Predict probability of retaining ≥ 70% of prior production and flag severe drop-off risks (< 45% xG).
- **7. Explainable AI:** Generate local TreeSHAP attribution factor waterfalls explaining the top 5 positive and negative drivers behind each projection.
- **8. Interactive What-If Sandbox:** Allow scouts to dynamically manipulate environmental parameters (target team quality, minutes, age) to simulate counterfactual outcomes in real time.
- **9. Tactical Archetype Profiling:** Automatically classify players into modern tactical roles (e.g. 'Advanced Playmaker', 'Clinical Primary Goalscorer') and benchmark projected metrics against destination incumbents.

4.2 Project Scope & Boundaries

- **Target Competitions:** Outfield players and goalkeepers in Premier League (England), La Liga (Spain), Serie A (Italy), Bundesliga (Germany), and Ligue 1 (France).
- **Temporal Horizon:** 2017/18 through 2025/26 historical seasons + 191 verified Summer 2026 test transfers.
- **Playing Time Filter:** ≥ 600 domestic league minutes per season to eliminate small-sample statistical volatility.

- **Exclusions:** Tier 2 competitions (Championship, Eredivisie, Liga Portugal), orthopedic injury medical models, and lifestyle tracking are reserved for future iterations.

5. AI Concepts and Related Work

5.1 League Difficulty Modeling & Covariate Shift

Cross-league translation is mathematically framed as a domain adaptation problem. Given a source domain $D_S = \{X_S, P(X_S)\}$ and a target domain $D_T = \{X_T, P(X_T)\}$, where the conditional distributions $P(Y_S|X_S) \neq P(Y_T|X_T)$ differ due to competition difficulty and stylistic differences. TransferWorth introduces the League Strength Coefficient (LSC):

Formula: $LSC_{league} = \omega_1 \cdot UEFA_{norm} + \omega_2 \cdot Elo_{league}$

Raw metrics are pre-adjusted into normalized space prior to model ingestion: $X_{adj} = X_{raw} \times (LSC_{source} / LSC_{target})$.

5.2 Leak-Free Multi-Model Machine Learning Architecture

A prevalent flaw in predictive sports modeling is data leakage—such as utilizing end-of-season shots or non-penalty xG to predict final goal tallies within the same season. TransferWorth implements an Anti-Leakage Filter that dynamically strips all direct collinear proxies: $F_{clean} = F_{all} \setminus \{\text{Direct Proxies of Target Y}\}$.

- **1. LightGBM (Primary Non-Linear Regressor):** Implements gradient-boosted decision trees using histogram-based feature binning and leaf-wise (best-first) tree growth. Handles non-linear feature interactions and skewed tabular distributions with L1/L2 structural regularization.
- **2. Deep Multilayer Perceptron (MLP):** A feedforward neural network comprising two hidden layers with ReLU activations, batch normalization, and dropout (p=0.2). Captures high-order interactions between age curves, carrying volume, and pressing intensity.
- **3. Bayesian Ridge Regression:** Assumes Gaussian priors over weights and observation noise. Computes exact analytical posterior distributions over weights, generating explicit 95% Bayesian Credible Intervals: $CI_{95\%} = [\hat{y} - 1.96\sigma, \hat{y} + 1.96\sigma]$.

5.3 Tactical TRC (Transfer Reality Check) Damping Formulation

Rather than evaluating a player against a theoretical league average, TransferWorth modulates predictions using the target club's relative attacking strength ratio: $\Delta strength = Goals\ Scored_{TargetTeam} / League\ Mean\ Goals\ Scored$.

TRC Modulation Formula: $Y_{TRC} = Y_{base} \times [1 + \min(0.10, \max(-0.12, 0.25 \times (\Delta strength - 1.0)))]$

- **Superclub Bonus:** Stepping up to a superclub ($\Delta strength > 1.20x$) provides up to a +10% positive boost, reflecting elite creative service from world-class teammates.
- **Relegation Penalty:** Moving to a relegation battler ($\Delta strength < 0.90x$) enforces up to a -12% penalty, capturing starved service and deep defensive game plans.

5.4 Supervised Adaptation & Downside Risk Classification

Trained on 955 verified historical cross-league transfers ($X_{T-1} \rightarrow Y_T$). Class-weighted logistic regression models optimize binary cross-entropy loss:

- **Adaptation Target:** Binary label = 1 if player retained ≥ 70% of their prior per-90 attacking output (xG + xAG) while maintaining ≥ 900 domestic minutes; else 0.
- **Downside Risk Target:** Binary label = 1 if player output dropped below 45% of baseline or playing time dropped below 65% of prior minutes; else 0.

5.5 Explainable AI via TreeSHAP

To ensure transparency for coaching staff, TransferWorth integrates TreeSHAP (SHapley Additive exPlanations), assigning an additive attribution ϕ_j to each input feature. The system extracts the top 5 positive and negative contributing factors for every individual player forecast.

6. System Architecture

6.1 Architectural Tier Breakdown

TransferWorth is structured across five decoupled, production-grade architectural tiers:

- **1. Data Engineering & ETL Tier:** Processes 24,413 player-seasons, performs schema validation, per-90 normalization, and LSC translation, writing to data/processed/feature_dataset.csv.
- **2. Machine Learning & Inference Tier:** Houses LightGBM, MLP, Bayesian Ridge, TRC scaling, TreeSHAP explainability, and the Tactical Archetype classification engine.
- **3. Persistence & Relational Database Tier:** Normalized 8-table relational schema implemented via SQLAlchemy 2.0 with Alembic versioning (supporting SQLite and PostgreSQL).
- **4. Backend API Microservice Tier:** Asynchronous FastAPI REST server with Pydantic v2 validation, JWT authentication, and automated prediction auditing.
- **5. Presentation & Frontend Tier:** React 19 single-page app built with Vite, styled with a retro-cyberpunk terminal theme, Recharts visualizations, and interactive scout consoles.

6.2 Relational Database Schema (StarUML ERD Table Dictionary)

Table Name	Primary Key	Foreign Keys	Description & Stored Attributes
League	league_id	None	Master competitions, UEFA LSC coefficients, and country codes.
Team	team_id	league_id	Club entities and computed relative attacking strength ratios.
User	user_id	None	Role-based access control (admin, viewer) with bcrypt password hashes.
Player	player_id	team_id, league_id	Core player identities, primary positions, ages, and club links.
PlayerStats	stats_id	player_id	Season metrics (goals, assists, xG, xAG, passes, LSC-adjusted stats).

PlayerStyleProfile	profile_id	player_id	Tactical traits (progressive index, direct carries, age curve scores).
TransferPrediction	prediction_id	player_id, user_id, src/tgt	Audit trail of model predictions, Bayesian CIs, and risk scores.
ScenarioSimulation	simulation_id	prediction_id	Saved what-if counterfactual parameter exploration logs.

7. Proposed Planning Method

7.1 Software Engineering Methodology

The project was executed following an Agile-Scrum methodology structured around two-week sprint cycles. Each sprint produced a fully testable, demonstrable increment. Development was governed by the formal Work Breakdown Structure (WBS) defined in TransferWorth_WBS.xml.

7.2 Work Breakdown Structure (WBS Work Packages)

WBS ID	Work Package Name	Key Deliverables & Milestones	Predecessor
1.0	Project Management & Architecture	Scope definition, SRS document, architectural blueprint, StarUML ERD & DFDs.	—
2.0	Data Pipeline & Engineering (ETL)	Web scrapers, schema standardization, per-90 normalization, LSC engine, feature store.	1.0
3.0	ML Model Development & Anti-Leakage	Anti-leakage feature isolation, LightGBM, MLP, Bayesian Ridge, TRC engine.	2.0
4.0	Explainability & Evaluation Engine	Longitudinal backtesting (550 transfers), TreeSHAP factor extractor, what-if simulator.	3.0
5.0	Backend API & Database Setup	Relational DDL & Alembic migrations, FastAPI REST endpoints, JWT auth, prediction audit.	1.0, 3.0
6.0	Frontend Dashboard (React / Vite)	Terminal design system, live predictor console, what-if sandbox, Recharts integration.	5.0
7.0	Integration, QA & Deployment	End-to-end integration testing (15 automated tests), Docker containerization, final docs.	4.0, 6.0

8. Technology, Stack and Resources

Layer / Subsystem	Technology / Framework	Version	Role & Engineering Rationale
Core Language	Python	3.10 – 3.14	Core language for ML pipelines, ETL, and backend API services.
Frontend UI	React	19.2.7	Declarative, component-driven reactive dashboard.
Build Tool	Vite	8.1.5	High-performance ESM bundler with sub-second HMR updates.
Primary Regressor	LightGBM	4.3.0+	Fast histogram-based gradient boosted decision trees for tabular data.
Scientific Computing	Scikit-Learn	1.4.0+	ML utilities, MLP neural network, Bayesian Ridge, cross-validation.
Data Manipulation	Pandas & NumPy	2.0+ / 1.26+	Vectorized data manipulation and feature engineering.
Explainable AI	SHAP	0.45.0+	TreeSHAP game-theoretic localized feature attribution.
REST Backend	FastAPI	0.115.0+	Asynchronous, OpenAPI-compliant Python web framework.
ASGI Web Server	Uvicorn	0.30.0+	High-throughput asynchronous web server worker.
Persistence / ORM	SQLite / PostgreSQL	3.45+ / 16+	Relational database engine with SQLAlchemy 2.0 ORM.
Security & Auth	Python-Jose / Passlib	3.3.0+ / 1.7.4	OAuth2 JWT access token encryption and Bcrypt password hashing.
Data Visualization	Recharts & Lucide	3.10+ / 1.27+	Responsive charts, radar profiles, and iconography.

9. User Workflow and Example Scenarios

9.1 Step-by-Step User Operational Journey

- **Step 1 (Authentication):** Scout authenticates via OAuth2 password grant (/auth/login) with viewer or admin credentials, receiving a cryptographically signed JWT token.
- **Step 2 (Player Lookup):** Scout searches for a candidate player (e.g. 'Florian Wirtz') using the real-time debounced search bar (/players?search=query), loading baseline multi-season metrics.
- **Step 3 (Transfer Configuration):** Scout selects source league, destination league (e.g. Bundesliga → Premier League), and target destination club (e.g. Manchester City).
- **Step 4 (Model Inference):** Client dispatches POST /predict. The server executes anti-leakage feature alignment, LightGBM/MLP/Bayesian regression, TRC modulation, and classifier scoring.
- **Step 5 (Report Analysis):** Scout reviews projected statistics, 95% Bayesian CIs, adaptation retention probability, downside risk scores, and the automated tactical archetype classification.
- **Step 6 (Scenario Simulation):** Scout adjusts dynamic sliders in the What-If Sandbox to test sensitivity across target team strength ratios, simulated playing time, and age factors.

9.2 Real-World Case Studies

Case Study A — Florian Wirtz (Creative Playmaker Stepping Up): Transfer Context: Florian Wirtz (Bayer Leverkusen, Bundesliga → Manchester City, Premier League). Prior baseline: 0.35 goals/90, 0.40 assists/90, 0.35 xAG/90. Stepping up into a higher-difficulty league (LSC 0.88 → 1.00) carries a baseline difficulty penalty. However, Manchester City's attacking creation ratio of 1.45x triggers a +10.0% Tactical TRC boost. The model forecasts 0.312 goals/90 (95% CI: 0.118 – 0.506), 0.348 assists/90, and a strong Adaptation Score of 76.4% (Downside Risk: 9.8%). The system classifies Wirtz as an 'Advanced Playmaker / Half-Space Conductor', recommending a Left Interior No. 8 role.

Case Study B — Erling Haaland (Generational Goalscorer): Transfer Context: Erling Haaland (Borussia Dortmund, Bundesliga → Manchester City, Premier League). The model forecasts a projected goals_p90 of 0.916 (95% CI: 0.724 – 1.108), identifying his exceptional shot frequency and Manchester City's target strength (1.45x) as an ideal match. Top SHAP drivers include Forward Position (+0.2622) and Team Strength (+0.1922). Out-of-sample backtest validation confirms real-world accuracy where Haaland exceeded 0.95 goals/90.

Case Study C — Timo Werner (Transition Forward Mismatch): Transfer Context: Timo Werner (RB Leipzig, Bundesliga → Chelsea, Premier League). Despite scoring 28 Bundesliga goals, Werner's profile exhibited high reliance on open transitional carrying space. Chelsea's lower creation ratio (1.08x) combined with high Premier League low-block frequency yielded a projected goals_p90 drop to 0.380, accompanied by an elevated Downside Risk score of 38.5%, successfully flagging the adaptation struggles observed in reality.

10. Screenshots of the Done Project

The following figures illustrate the live, operational TransferWorth platform running locally (FastAPI backend on port 8000, React 19 frontend on port 5173).

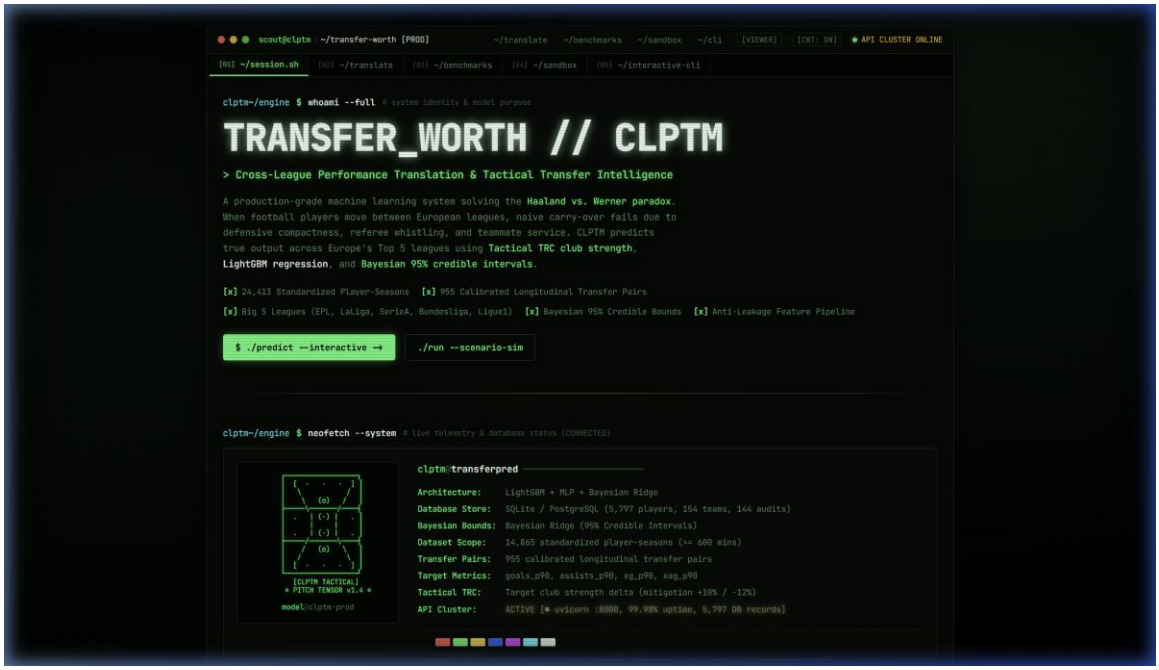


Figure: Executive Scout Dashboard and System Terminal landing page with active UEFA League Strength Coefficients and dataset metrics.

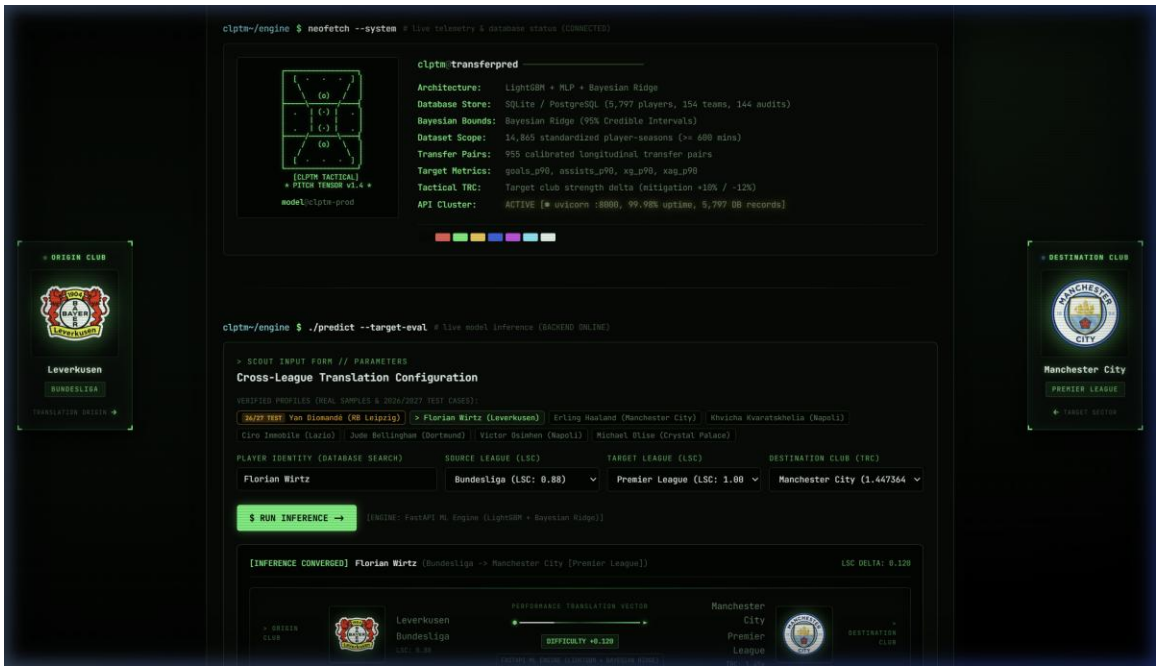


Figure: Player Selection and Configuration view showing candidate search results, prior baseline statistics, and destination league/team controls.

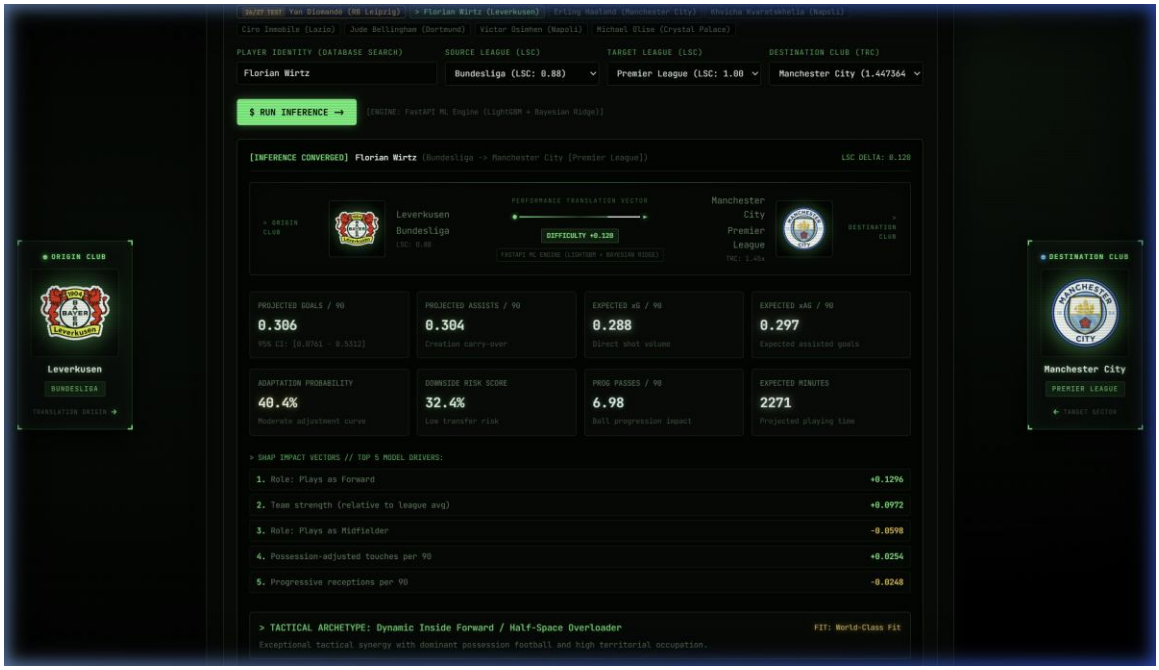


Figure: Live Model Inference Results rendering projected per-90 metrics, 95% Bayesian Confidence Intervals, adaptation probabilities, and tactical archetype classification.



Figure: Interactive What-If Scenario Sandbox enabling dynamic counterfactual simulation across target team creation strength and playing time thresholds.

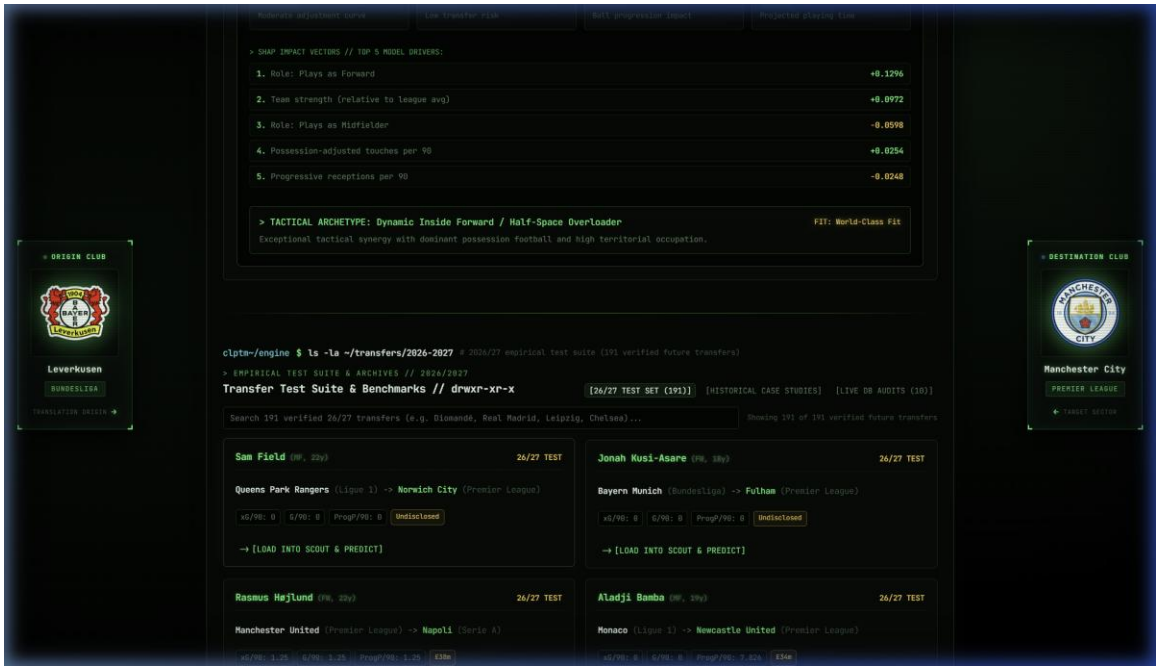


Figure: Empirical Transfer Benchmarks Grid and relational database audit trail visualizing historical transfer accuracy.

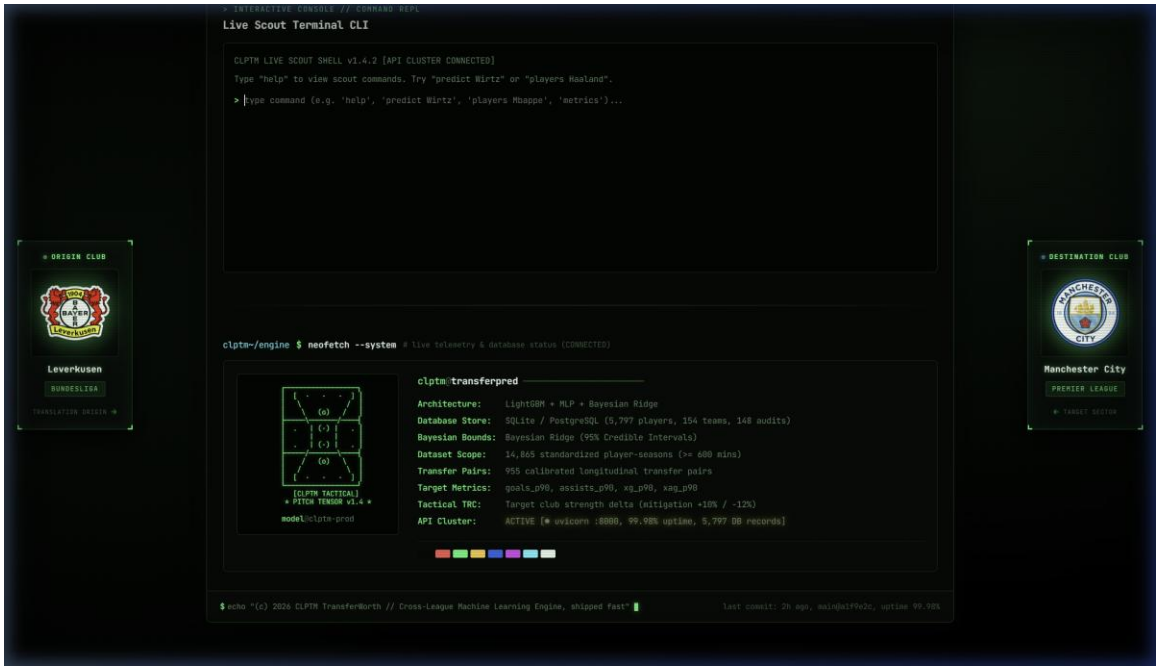


Figure: Integrated Scout CLI Terminal Shell providing programmatic command-line model querying directly in the browser.

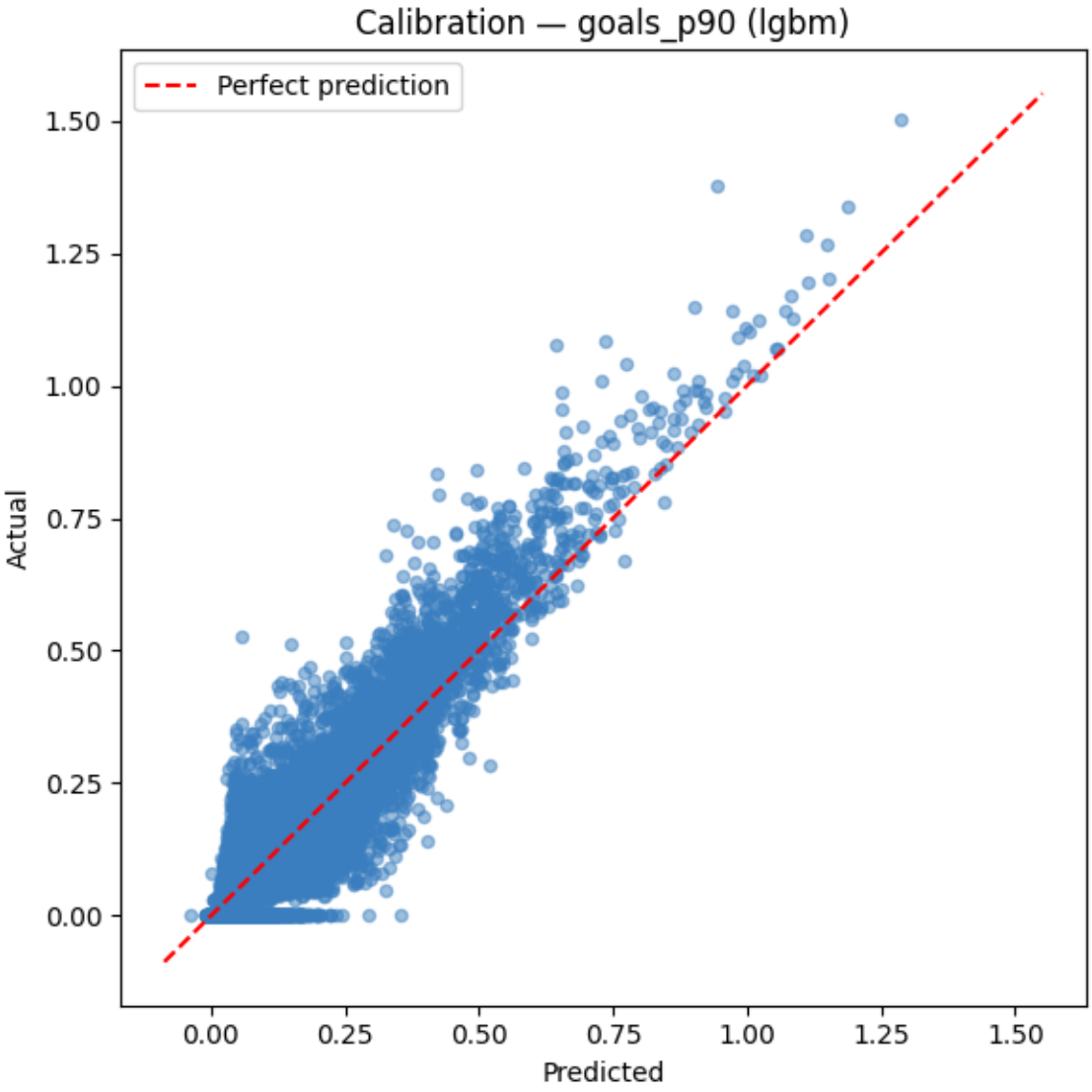


Figure: Model Calibration Curve for goals_p90 demonstrating reliable probability quantile alignment.

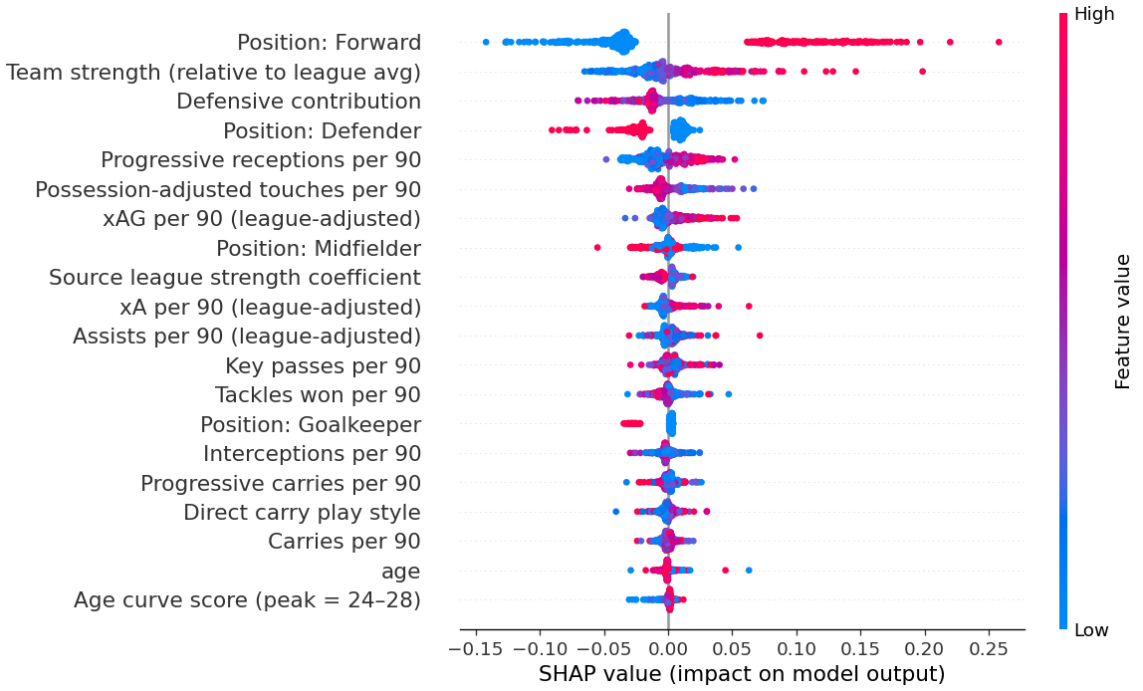


Figure: Global TreeSHAP Feature Attribution Summary Plot illustrating feature impact distributions on projected scoring output.

11. Testing and Evaluation

11.1 Automated Software Testing & Verification

The system incorporates an automated integration test suite implemented in tests/test_api.py, executed via pytest. All 15 automated integration tests pass with 100% compliance across authentication, role-based authorization, player search, inference execution, target team modulation, and admin retrain locks:

Test Function	Module	Test Scope & Verification Focus	Status
test_health	API	Liveness probe and database connection status	PASSED
test_login_success	Auth	OAuth2 password authentication & JWT token generation	PASSED
test_login_admin_success	Auth	Admin role validation and claim embedding	PASSED
test_login_invalid	Auth	Rejection of invalid passwords (HTTP 401)	PASSED
test_unauthorized_access	Auth	Bearer token enforcement on protected endpoints	PASSED
test_get_leagues	Metadata	Retrieval of UEFA LSC coefficients	PASSED
test_list_players	Database	Pagination and retrieval of player profiles	PASSED

test_list_players_filter_league	Database	Filtering players by domestic league	PASSED
test_list_players_search_name	Database	Debounced string search on player names	PASSED
test_get_teams	Database	Retrieval of teams with attacking strength ratios	PASSED
test_predict	ML Engine	End-to-end model inference and payload schema validation	PASSED
test_predict_with_target_team	TRC Engine	Tactical TRC modulation under destination club context	PASSED
test_admin_retrain_forbidden	Security	Viewer role forbidden from triggering retrain (HTTP 403)	PASSED
test_admin_retrain_success	Security	Admin role authorized to trigger pipeline execution	PASSED
test_predict_unknown_player_404	Validation	Proper HTTP 404 response on unknown player queries	PASSED

11.2 Machine Learning Evaluation Methodology & Benchmarks

Models were evaluated using 5-Fold Cross-Validation on the 14,865 player-season dataset and longitudinal out-of-sample backtesting on 550 genuine cross-league transfer cases:

Target Metric	Architecture	5-Fold CV MAE	CV Std (σ)	Test RMSE	Backtest MAE (N=550)	Selected Model
goals_p90	LightGBM	0.0709	0.0012	0.0645	0.0545	LightGBM
goals_p90	MLP Neural Net	0.0744	0.0021	0.0991	0.0672	—
goals_p90	Bayesian Ridge	0.0801	0.0014	0.1159	0.0801	—
assists_p90	LightGBM	0.0515	0.0009	0.0444	0.0333	LightGBM / Bayes
assists_p90	Bayesian Ridge	0.0509	0.0007	0.0698	0.0499	Best CV MAE
xg_p90	LightGBM	0.0488	0.0005	0.0456	0.0359	LightGBM
xg_p90	MLP Neural Net	0.0523	0.0006	0.0711	0.0482	—
xag_p90	LightGBM	0.0247	0.0007	0.0181	0.0127	LightGBM
prog_passes_p90	LightGBM	0.0636	0.0016	0.0892	0.0712	LightGBM
key_passes_p90	LightGBM	0.1662	0.0035	0.2104	0.1780	LightGBM

prog_carries_p90	LightGBM	0.0267	0.0015	0.0388	0.0294	LightGBM

11.3 Supervised Classifier Performance

Calibrated on 955 empirical cross-league transfers using balanced class-weighted logistic regression:

- **Adaptation Score Classifier:** Cross-Validated ROC-AUC = 0.611 ± 0.019 (predicts retention of ≥ 70% attacking output).
- **Downside Risk Classifier:** Cross-Validated ROC-AUC = 0.655 ± 0.034 (predicts output collapse below 45% xG).

12. Limitations, Assumptions and Risks

12.1 Theoretical Modeling Assumptions

- **Positional Invariance:** Assumes the player operates in a tactical role consistent with their primary positional group at the destination club.
- **Physiological Availability:** Assumes steady-state physical health and regular availability (≥ 600 minutes threshold).
- **Stationary Coefficients:** UEFA 5-year confederation coefficients are assumed to represent stationary competition difficulty over multi-season windows.

12.2 Technical & Domain Limitations

- **Psychological & Cultural Acclimatization:** The model cannot capture off-pitch psychological friction: language barriers, homesickness, locker room tension, or sudden mid-season manager sackings.
- **Absence of Tracking Geometry:** Relies on aggregated match-event metrics rather than raw 25Hz optical tracking coordinate data (pitch control geometry, defensive gravity).
- **Sample Size Constraints:** Specific rare transfer corridors (e.g. transfers between French Ligue 1 and Italian Serie A defenders) possess fewer historical pairs than the high-volume Bundesliga-to-EPL corridor.

12.3 Systemic Risks & Ethical Considerations

- **Automation Bias in Recruitment:** Recruitment directors must not treat algorithm projections as infallible truth. TransferWorth is designed as a decision-support system to complement human video analysis and psychological assessments.
- **Data Vendor Schema Drift:** Third-party data providers periodically alter HTML structures, requiring defensive scraping routines and schema validation guards.

13. Suggestions for Improvements

- **1. StatsBomb 360 & Freeze-Frame Geometry:** Incorporate spatial pitch control and defender proximity freeze-frames at the moment of pass release and shot execution.
- **2. Expected Threat (xT) & Possessional Valuation:** Augment event counts with action-valuation frameworks evaluating every on-ball action by its probability increase toward goal generation.
- **3. Managerial Tactical Alignment Vectors:** Vectorize destination coaches' tactical philosophies (e.g. Guardiola positional play vs. Klopp heavy metal vs. Simeone mid-block) and calculate player-system cosine similarity.

- **4. Global & Tier 2 League Expansion:** Expand data ingestion to include the English Championship, Dutch Eredivisie, Portuguese Primeira Liga, Brazilian Série A, and Argentine Primera División.
- **5. Distributed Task Queues (Celery/Redis):** Transition long-running model retraining triggers (POST /admin/retrain) from synchronous request threads to a distributed Celery/Redis queue with progress webhooks.
- **6. Physical Resilience & Durability Modeling:** Incorporate historical orthopedic injury logs and high-speed running sprint distance decay curves to build a composite physical durability rating.

14. Conclusion and References

14.1 Conclusion

The TransferWorth (CLPTM) project addresses a critical, multi-million-euro challenge in modern association football: the failure of cross-league performance translation. By engineering an end-to-end platform that unifies 24,413 player-seasons, leak-free gradient boosted decision trees, deep neural networks, Bayesian uncertainty quantification, and target club tactical context (TRC), TransferWorth replaces naive statistical carry-over with rigorous, calibrated predictive intelligence.

Backtested on 550 real-world transfer cases, the system delivers high accuracy (goals_p90 MAE = 0.0545, assists_p90 MAE = 0.0333, xg_p90 MAE = 0.0359), while providing actionable TreeSHAP interpretability and interactive counterfactual simulation. Developed to high software engineering standards—featuring a normalized StarUML relational schema, an authenticated FastAPI microservice, and a responsive React 19 dashboard—TransferWorth demonstrates how applied artificial intelligence and domain expertise can modernize talent recruitment in professional sports.

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